

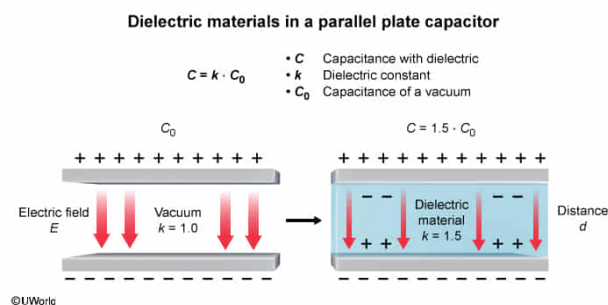
A dielectric material is introduced into the empty space between the two plates of a parallel plate capacitor. How will the capacitance (C) change if the value of the dielectric constant (κ) for the material is 1.5?

- ☐ A. C will decrease by 33%. [9%]
- ☐ B. C will not change. [8%]
- ✓ ☒ C. C will increase by 50%. [78%]
- ☐ D. C will increase by 100%. [3%]

Correct

78% answered correctly

Explanation:



A **capacitor** is a device that stores electrical charge between equally but oppositely charged conductive surfaces (plates) separated by a fixed distance (d). Capacitor strength is determined by the quantity of charge (Q) stored on each plate relative to the voltage (V), or attractive electric potential energy, this separation of charges generates. Capacitor strength is quantified by **capacitance (C)**, a variable generally given in units of farads (F) and calculated using:

$$C = \frac{Q}{V}$$

In the case of a simple **parallel plate capacitor**, capacitance is determined largely by physical dimensions and plate material. However, **dielectrics** can be inserted between capacitor plates such that the characteristics of a capacitor are altered. Dielectrics are a class of polarizable materials in which electric dipoles can be induced. When a dielectric material is inserted between capacitor plates, the electric field present between these plates causes the dipoles induced in the dielectric material to align, *decreasing both* electric field strength and voltage. Due to this insulating effect, dielectric substances *increase* capacitance.

Accordingly, the **dielectric constant (κ)** reflects the ability of dielectrics to increase the functional capacitance beyond the capacitance in a vacuum (C_0), defined as the absence of any material. This relationship is expressed as:

$$C = \kappa \cdot C_0$$

In this question, a dielectric material with a dielectric constant of 1.5 is introduced into the previously empty space of a parallel plate capacitor under vacuum. Applying the above equation, the new capacitance following the addition of the dielectric material can be calculated as:

$$C = 1.5 \cdot C_0$$

Therefore, introducing a dielectric with a dielectric constant of 1.5 will increase capacitance by 50% **(Choice C)**.

(Choices A and B) Both a vacuum and dielectric materials have κ values that are greater than 1. As such, introducing any material with $\kappa \geq 1$ *must* increase capacitance.

(Choice D) A material with a κ value of 2 would have to be introduced between capacitor plates for capacitance to increase 100%. However, the dielectric used in this question has a κ value of only 1.5.

Educational objective:

A dielectric can be introduced between the plates of a parallel plate capacitor to cause an increase in capacitance. The dielectric constant κ measures the ability of a dielectric material to increase capacitance.

Time spent:0 sec

Subject:

Physics

QID:402054

System:

Electrostatics and Circuits

The resistance of an object acting as a resistor is directly proportional to its:

- I. resistivity.
 - II. cross-sectional area.
 - III. length.
- ☐ A. I only [5%]
☐ B. II only [2%]
☐ C. I and II only [20%]
✓ ☒ D. I and III only [71%]

Correct

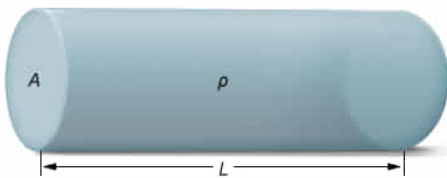
71% answered correctly

Explanation:

Resistance of a resistor

$$R = \rho \cdot \frac{L}{A}$$

- R Resistance
- ρ Resistivity
- L Length
- A Cross-sectional area



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In electronics, **resistance (R)** refers to the capacity of an object acting as a resistor to oppose the movement of charge (ie, current). Resistance is expressed by [Ohm's law](#) as the ratio of the voltage (V) applied across the resistor relative to the current (I) flowing through it.

The resistance of a given object (eg, a wire within a circuit) quantifies the relative difficulty by which charge moves from one end of the object to the other. As a result, *resistance* increases in [direct proportion](#) to *length* (L) because increased length forces charge to travel through a greater distance of resistant material (**Number III**):

$$R \propto L$$

Conversely, resistance *decreases* as the cross-sectional area (A) increases because additional cross-sectional area provides more parallel channels through which electrical charge can flow. As a result, resistance is [inversely proportional](#) to cross-sectional area (**Number II**):

$$R \propto \frac{1}{A}$$

As another variable that influences resistance, **resistivity (ρ)** can be used to quantify the differential resistance of materials. Most often reported in ohm-meters ($\Omega\cdot m$), resistivity is the inherent resistance per unit length for a uniform cross-section of the object and is expressed mathematically as:

$$\rho = R \cdot \frac{A}{L}$$

The resistance of two objects with equal physical dimensions can differ based on the material that constitutes the resistors. Accordingly, the resistance of an object made from a high-resistivity material will be greater than the resistance of an object made from a low-resistivity material of equal physical dimensions. Therefore, *resistivity* is **directly proportional** to *resistance (Number 1)*. Consequently, the equation for resistivity can be rearranged to yield an overall expression for resistance:

$$R = \rho \cdot \frac{L}{A}$$

Educational objective:

The resistance of an object is directly proportional to its resistivity and length but inversely proportional to its cross-sectional area. Resistivity assesses the intrinsic resistance within a defined volume of a material.

Time spent:0 sec

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